

Xai

Intelligence Workspace

Product Documentation

July 2026

From raw data !' structured intelligence !' actionable insight !' AI automations.

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1. Product Overview

Xai is a single-page product experience that walks through the full data-to-decision loop. The narrative arc moves from raw data ingestion, through structured intelligence and actionable insights, to AI-powered automation — all rendered as an immersive, scrollable web experience.

The core metaphor: data enters as chaos, is shaped into structure by models, surfaces as ranked insight, and triggers automated actions. Each scroll section represents one stage in this transformation.

Built as a front-end prototype with a PostgreSQL-backed access request flow, the experience demonstrates production-quality engineering: Three.js 3D visualizations, responsive design, dual theme support, smooth scroll, and accessible form interactions.

Key narrative pillars:

- Raw data becomes intelligence (Hero section)
- Three stages from noise to signal (Insight Flow)
- A calm surface for complex intelligence (Dashboard)
- One core that reorganizes itself (Signature)
- Insight becomes action, on its own (Automations)
- Built for the full loop (Pillars + Metrics)

2. Tech Stack

Framework: Next.js 16 (App Router)

Language: TypeScript

Styling: Tailwind CSS v4 + CSS custom properties (oklch)

Animation: Framer Motion + Lenis (smooth scroll)

3D: Three.js via @react-three/fiber + @react-three/drei

Charts: Recharts

State: Zustand (auth store with localStorage persist)

Database: PostgreSQL + Prisma ORM

Forms: Controlled forms + Zod v4 validation

Icons: Lucide React

Font: Geist (sans + mono via next/font)

3. Page Architecture & Routes

/ (landing)

8 sections: Hero, Narrative, Insight Flow, Dashboard, Signature, Automations, Pillars, Metrics

/signin

Sign-in with demo credentials (admin / member) + demo fill buttons

/request-access

5-phase state machine: form !' submitting !' success / duplicate / error

/admin

Admin dashboard: overview, requests, users, activity tabs — role-gated

/api

Health check endpoint

404

Custom not-found page with motion animations

Auth is handled client-side via Zustand with localStorage persist. Two hard-coded demo accounts are configurable via env vars:

admin@xai.app / xai-demo (role: admin)
member@xai.app / xai-demo (role: member)

4. Section-by-Section Breakdown

Hero (#top)

Three.js particle field (1800 particles on desktop, 800 on mobile) that auto-oscillates between a chaotic 'raw data' cloud and a structured Fibonacci sphere. GLSL shader morphs position, color, and size. Cursor parallax drives group rotation and camera dolly. A CSS-based aurora + grid backdrop provides atmosphere.

Narrative Band

Four-card rail showing the transformation moves: Raw Data !' Structured Intelligence !' Actionable Insight !' AI Automation. Cards are arranged in a 4-column grid with a connecting gradient line animation. Ghost number index for depth effect. Glass-morphism cards with lift-on-hover.

Insight Flow (#flow)

Three scroll-reveal stages (Ingest !' Analyze !' Generate). Each stage pairs text content with a hand-built SVG visualization. Ingest shows multiple source streams converging into a unified schema. Analyze shows a 3-layer neural network diagram with signal path. Generate shows a grid heatmap + insight cards. No GSAP — uses framer-motion whileInView.

Dashboard (#workspace)

Full mock product UI embedded as a scroll-reveal perspective card. Includes sidebar nav with avatar, tabbed top navigation with spring-animated layoutId pill, 4 KPI cards with sparklines, area chart (insight velocity), bar chart (throughput), insights table, models panel with accuracy bars, sources panel with health meters, and automations panel. All powered by Recharts with theme-aware colors via useThemeColors hook.

Signature (#core)

Three.js 3D 'intelligence core': a distorted icosahedron (MeshDistortMaterial) inside a wireframe shell, surrounded by 54 orbiting nodes (28 on mobile) that reorganize from chaos into planar rings. Two torus rings rotate at different angles. Dual point lights (emerald + amber). Camera dolly transitions from wide to close.

Automations (#automations)

Trigger-action cards with success rate meters. Each card shows: name, trigger condition (if), action (then), run count, success percentage. Progress bars animate on scroll. Pipeline rail with sweeping gradient light. 'New automation' dashed-border composer card. 2-column grid on desktop.

Pillars + Metrics

Pillars: three cards (Ingest, Analyze, Automate) with hover glow effects, metric badges, links to flow section. Metrics Band: four animated KPI numbers (9 sources, 1,284 insights/wk, 248ms p50, 37 automations) with baseline progress bars. Alternating accent colors.

5. Three.js Interactions

Both 3D scenes are lazy-loaded via next/dynamic with `ssr: false` and include a pulsing fallback placeholder.

Hero Canvas

1800 particles (800 mobile) use a custom ShaderMaterial with GLSL vertex/fragment shaders. Two states: 'chaos' (random distribution) and 'structure' (Fibonacci sphere). The morph progress is driven by a $\sin$ *smoothstep oscillator (period ~15.7s). Uniforms include `uProgress`, `uTime`, `uPixelRatio`, and theme-aware colors. Cursor parallax via `useThree().pointer`. Camera dolly adjusts with progress and viewport.

Signature Canvas

A distorted icosahedron (`MeshDistortMaterial`) with animated `distort` and `emissiveIntensity`, nested inside a wireframe icosahedron, surrounded by 54 orbit nodes (28 mobile) that form 3 tiers. Two torus rings rotate at different Euler angles. Dual point lights (emerald + amber). Orbit nodes use `lerp` to transition from random positions to structured planar rings. Camera dolly responsive.

6. Data Flow & Server Actions

The only server-side data flow is the access request form:

- Client form collects name, email, company, role, use case
- Server Action (submitRequest) validates with Zod v4 schema
- Prisma writes to PostgreSQL AccessRequest table
- Unique email constraint catches duplicates (P2002)
- Typed result drives a 5-phase animated state machine (form/submitting/success/duplicate/error)

Auth is entirely client-side:

- Zustand store with persist middleware (localStorage)
- Two hard-coded demo accounts with env-var overridable credentials
- signIn returns { ok, error } — no real server verification
- Admin page uses useSyncExternalStore for SSR-safe hydration check

7. UI Component Decomposition

`role-select.tsx`

Animated custom dropdown with framer-motion, keyboard navigation (arrow keys, Enter, Escape, Space), WAI-ARIA combobox pattern, `scrollIntoView` for focused item, `layoutId` checkmark animation

`toast.tsx` / `toaster.tsx`

Radix UI toast primitives with CVA variants, used by `layout.tsx`

`scroll-progress.tsx`

Fixed top gradient bar that scales X with scroll position via `useSpring`

`scroll-to-top.tsx`

RAF-throttled scroll listener, `AnimatePresence` fade/scale, appears after 600px

`smooth-scroll.tsx`

Lenis instance with `prefers-reduced-motion` guard, configurable duration/easing

`site-nav.tsx`

Fixed nav with blur backdrop, scroll-aware border transition, mobile hamburger with `AnimatePresence` menu, body overflow lock

`site-footer.tsx`

CTA band with glass-strong card, floating ping indicator, copyright with dynamic year

`premium-section-label.tsx`

Parses 'NN · Title' pattern, renders 5 unique hand-crafted SVG geometric icons per section

`scene-3d-background.tsx`

CSS-only 3D depth: perspective grid floor, two floating orbs with drift animations, cursor parallax CSS vars, `THREE.Clock` warning suppression

8. Design System & Theme

Light and dark themes defined in oklch color space with a unified token set. Over 80 CSS custom properties for colors, radii, surfaces, and data visualization.

Core tokens (shared):

- --background, --foreground
- --accent (emerald), --accent-warm (amber)
- --card, --muted-foreground, --border, --ring
- --hairline, --surface-1/2/3 (tint system)
- --chart-1 through --chart-5 (data viz)
- --sidebar-* tokens
- --radius (0.75rem base)

Key utility classes:

- .glass / .glass-strong — backdrop-blur card surfaces
- .lift — hover translateY with accent border glow
- .edge-glow — pseudo-element gradient border overlay
- .text-gradient-accent — emerald !' amber gradient text
- .bg-grid / .bg-grid-fine — theme-aware grid patterns
- .bg-sec-{narrative|flow|workspace|core|automations|pillars|metrics} — per-section mesh gradients
- .bg-grain — SVG fractal noise overlay for tactile texture
- .auth-input — form input with focus accent border
- .text-display-{xl|lg|md|sm} — fluid clamp() typography

Section backgrounds use distinct radial gradient compositions so each section feels unique while maintaining visual cohesion. A grain overlay (SVG feTurbulence) adds tactile texture to every section. Shared easing curves across all animations.

9. Deployment (Netlify)

The project is configured for Netlify deployment:

- Build command: `npm run build`
- Publish directory: `.next`
- SPA redirects via `netlify.toml`
- Security headers (X-Frame-Options, HSTS, CSP) via `_headers` and `netlify.toml`
- `DATABASE_URL` must be set as env var in Netlify dashboard

10. Project Structure

Source tree (35 source files, ~5000 lines):

```
src/
app/
  layout.tsx      Root layout — ThemeProvider, Lenis, nav/footer, Toaster, 3D bg
  page.tsx        Landing — dynamic imports for 4 heavy sections
  globals.css     Dual-theme (oklch), glass/depth design system (588 lines)
  not-found.tsx   404 page with motion animations
  signin/page.tsx Sign-in form, demo credential fill buttons
  request-access/page.tsx 5-phase state machine (431 lines)
  admin/page.tsx   Admin dashboard — 4 tabs, role-gated (450 lines)
  actions/request-access.ts Server Action — Zod v4 + Prisma write
  api/route.ts     Health check
components/
  site/ (10 files) Hero, Signature, InsightFlow, DashboardPreview,
                  Automations, LandingSections, SiteNav, SiteFooter,
                  ScrollProgress, ScrollToTop, SmoothScroll, Scene3DBG
  three/ (2 files) HeroCanvas (GLSL particle morph), SignatureCanvas
  ui/ (3 files)    Toast, Toaster, custom/RoleSelect
hooks/ (1 file)   use-toast — reducer pattern toast state machine
lib/ (6 files)    motion (easing), mock-data, utils (cn), auth-store
                  use-theme-colors (Recharts sync), db (Prisma singleton)
prisma/
  schema.prisma   AccessRequest model (PostgreSQL)
public/
  favicon.ico, favicon.svg, site.webmanifest, robots.txt, _headers
```

Total: 35 source files | ~5000 lines | 0 ESLint errors | 0 TypeScript errors